25. Extraction of zinc from zinc blende is achieved by

(A) electrolytic reduction

(B) roasting followed by reduction with carbon

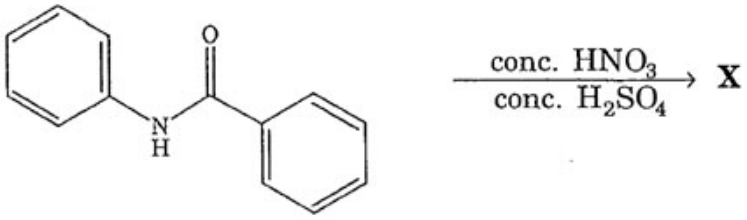
(C) roasting followed by reduction with another metal

(D) roasting followed by self-reduction

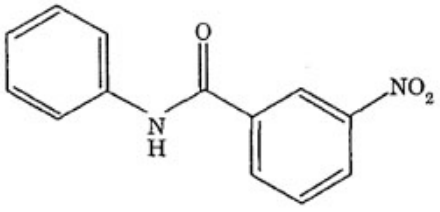
Answer ○ ● ○ ○

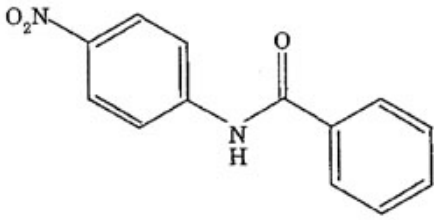
(A) (B) (C) (D)

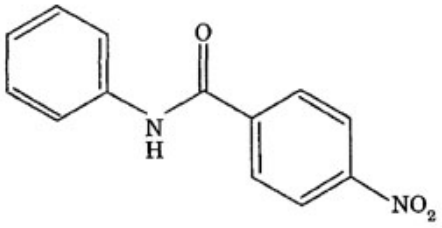
26. In the following reaction,

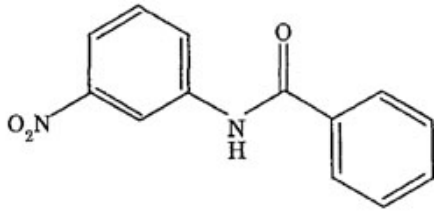


the structure of the major product '**X**' is

(A)


(B)


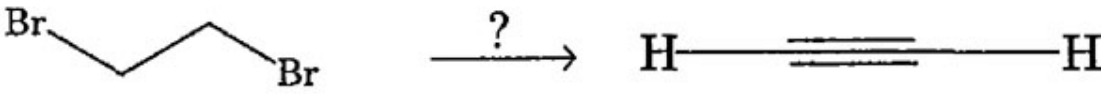
(C)


(D)


Answer ○ ● ○ ○

(A) (B) (C) (D)

27. The reagent(s) for the following conversion,



is/are

(A) alcoholic KOH

(B) alcoholic KOH followed by $NaNH_2$

(C) aqueous KOH followed by $NaNH_2$

(D) Zn/CH_3OH

Answer ○ ● ○ ○
(A) (B) (C) (D)

28. The number of structural isomers for C_6H_{14} is

(A) 3 (B) 4

(C) 5 (D) 6

Answer ○ ○ ● ○
(A) (B) (C) (D)

29. The percentage of p-character in the orbitals forming P–P bonds in P_4 is

(A) 25 (B) 33

(C) 50 (D) 75

Answer ○ ○ ○ ● (A) (B) (C) (D)

30. When 20 g of naphthoic acid ($C_{11}H_8O_2$) is dissolved in 50 g of benzene ($K_f = 1.72$ K kg mol^{-1}), a freezing point depression of 2 K is observed. The van't Hoff factor (i) is

(A) 0.5 (B) 1

(C) 2 (D) 3

Answer ● ○ ○ ○ (A) (B) (C) (D)

31. The value of $\log_{10}K$ for a reaction $A \rightleftharpoons B$ is

(Given : $\Delta_r H^\circ_{298K} = -54.07$ kJ mol^{-1}, $\Delta_r S^\circ_{298K} = 10$ $JK^{-1}mol^{-1}$ and $R = 8.314$ $JK^{-1}mol^{-1}$; $2.303 \times 8.314 \times 298 = 5705$)

(A) 5 (B) 10

(C) 95 (D) 100

Answer ○ ● ○ ○ (A) (B) (C) (D)

32. STATEMENT-1 : Boron always forms covalent bond.

because

STATEMENT-2 : The small size of B^{3+} favours formation of covalent bond.

(A) Statement-1 is True, Statement-2 is True; Statement-2 **is** a correct explanation for Statement-1

(B) Statement-1 is True, Statement-2 is True; Statement-2 **is NOT** a correct explanation for Statement-1

(C) Statement-1 is True, Statement-2 is False

(D) Statement-1 is False, Statement-2 is True

Answer ● ○ ○ ○

(A) (B) (C) (D)

33. STATEMENT-1 : In water, orthoboric acid behaves as a weak monobasic acid.

because

STATEMENT-2 : In water, orthoboric acid acts as a proton donor.

(A) Statement-1 is True, Statement-2 is True; Statement-2 **is** a correct explanation for Statement-1

(B) Statement-1 is True, Statement-2 is True; Statement-2 **is NOT** a correct explanation for Statement-1

(C) Statement-1 is True, Statement-2 is False

(D) Statement-1 is False, Statement-2 is True

Answer ○ ○ ● ○

(A) (B) (C) (D)

34. STATEMENT-1 : *p*–Hydroxybenzoic acid has a lower boiling point than *o*-hydroxybenzoic acid.

because

STATEMENT-2 : *o*–Hydroxybenzoic acid has intramolecular hydrogen bonding.

(A) Statement-1 is True, Statement-2 is True; Statement-2 **is** a correct explanation for Statement-1

(B) Statement-1 is True, Statement-2 is True; Statement-2 **is NOT** a correct explanation for Statement-1

(C) Statement-1 is True, Statement-2 is False

(D) Statement-1 is False, Statement-2 is True

Answer ○ ○ ○ ● (A) (B) (C) (D)

35. STATEMENT-1 : Micelles are formed by surfactant molecules above the critical micellar concentration (CMC).

because

STATEMENT-2 : The conductivity of a solution having surfactant molecules decreases sharply at the CMC.

(A) Statement-1 is True, Statement-2 is True; Statement-2 **is** a correct explanation for Statement-1

(B) Statement-1 is True, Statement-2 is True; Statement-2 **is NOT** a correct explanation for Statement-1

(C) Statement-1 is True, Statement-2 is False

(D) Statement-1 is False, Statement-2 is True

Answer ○ ● ○ ○ (A) (B) (C) (D)

36. Argon is used in arc welding because of its

(A) low reactivity with metal

(B) ability to lower the melting point of metal

(C) flammability

(D) high calorific value

Answer ● ○ ○ ○

(A) (B) (C) (D)

37. The structure of XeO_3 is

(A) linear (B) planar (C) pyramidal (D) T-shaped

Answer ○ ○ ● ○

(A) (B) (C) (D)

38. XeF_4 and XeF_6 are expected to be

(A) oxidizing (B) reducing (C) unreactive (D) strongly basic

Answer ● ○ ○ ○

(A) (B) (C) (D)

OR

○ ● ○ ○

(A) (B) (C) (D)

Chemical reactions involve interaction of atoms and molecules. A large number of atoms/molecules (approximately 6.023×10^{23}) are present in a few grams of any chemical compound varying with their atomic/molecular masses. To handle such large numbers conveniently, the mole concept was introduced. This concept has implications in diverse areas such as analytical chemistry, biochemistry, electrochemistry and radiochemistry. The following example illustrates a typical case, involving chemical/electrochemical reaction, which requires a clear understanding of the mole concept.

A 4.0 molar aqueous solution of NaCl is prepared and 500 mL of this solution is electrolysed. This leads to the evolution of chlorine gas at one of the electrodes (atomic mass : Na = 23, Hg = 200; 1 Faraday = 96500 coulombs).

39. The total number of moles of chlorine gas evolved is

(A) 0.5 (B) 1.0 (C) 2.0 (D) 3.0

Answer ○ ● ○ ○

(A) (B) (C) (D)

Chemical reactions involve interaction of atoms and molecules. A large number of atoms/molecules (approximately 6.023×10^{23}) are present in a few grams of any chemical compound varying with their atomic/molecular masses. To handle such large numbers conveniently, the mole concept was introduced. This concept has implications in diverse areas such as analytical chemistry, biochemistry, electrochemistry and radiochemistry. The following example illustrates a typical case, involving chemical/electrochemical reaction, which requires a clear understanding of the mole concept.

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40. If the cathode is a Hg electrode, the maximum weight (g) of amalgam formed from this solution is

(A) 200 (B) 225 (C) 400 (D) 446

Answer ○ (A) ○ (B) ○ (C) ● (D)

Chemical reactions involve interaction of atoms and molecules. A large number of atoms/molecules (approximately 6.023×10^{23}) are present in a few grams of any chemical compound varying with their atomic/molecular masses. To handle such large numbers conveniently, the mole concept was introduced. This concept has implications in diverse areas such as analytical chemistry, biochemistry, electrochemistry and radiochemistry. The following example illustrates a typical case, involving chemical/electrochemical reaction, which requires a clear understanding of the mole concept.

A 4.0 molar aqueous solution of NaCl is prepared and 500 mL of this solution is electrolysed. This leads to the evolution of chlorine gas at one of the electrodes (atomic mass : Na = 23, Hg = 200; 1 Faraday = 96500 coulombs).

41. The total charge (coulombs) required for complete electrolysis is

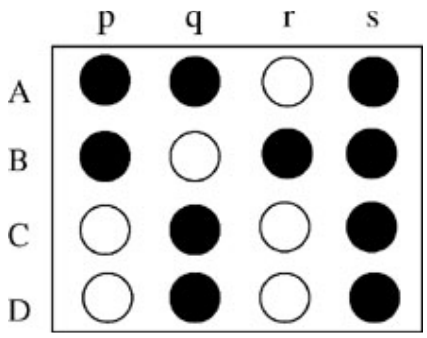
(A) 24125 (B) 48250 (C) 96500 (D) 193000

Answer ○ (A) ○ (B) ○ (C) ● (D)

42. Match the complexes in **Column I** with their properties listed in **Column II**. Indicate your answer by darkening the appropriate bubbles of the 4×4 matrix given in the ORS.

Column I	**Column II**
(A) $[Co(NH_3)_4(H_2O)_2]Cl_2$	(p) geometrical isomers
(B) $[Pt(NH_3)_2Cl_2]$	(q) paramagnetic
(C) $[Co(H_2O)_5Cl]Cl$	(r) diamagnetic
(D) $[Ni(H_2O)_6]Cl_2$	(s) metal ion with +2 oxidation state

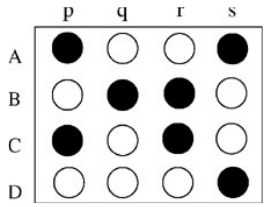
Answer



43. Match the chemical substances in **Column I** with type of polymers/type of bonds in **Column II**. Indicate your answer by darkening the appropriate bubbles of the 4×4 matrix given in the ORS.

Column I	**Column II**
(A) cellulose	(p) natural polymer
(B) nylon-6, 6	(q) synthetic polymer
(C) protein	(r) amide linkage
(D) sucrose	(s) glycoside linkage

Answer



44. Match gases under specified conditions listed in **Column I** with their properties/laws in **Column II**. Indicate your answer by darkening the appropriate bubbles of the 4×4 matrix given in the ORS.

Column I	**Column II**
(A) hydrogen gas (P = 200 atm, T = 273 K)	(p) compressibility factor $\neq$ 1
(B) hydrogen gas (P ~ 0, T = 273 K)	(q) attractive forces are dominant
(C) CO_2 (P = 1 atm, T = 273 K)	(r) PV = nRT
(D) real gas with very large molar volume	(s) P(V−nb) = nRT

Answer

